Chinese Sign Language Key Action Recognition Based on Extenics Immune Neural Network

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***Abstract*—In order to solve the problem of low accuracy of key action capture and feature recognition in Chinese sign language human-computer interaction, a large scale continuous sign language data set is established with Kinect equipment. By combining extension analysis theory with immune neural network (INN), extension immune detector is designed to locate the key actions of the sign language effectively. Hu invariant moment feature and Hamming rule are combined to improve immune neural network for accurate sign language image matching. The experimental results show that the sign language key motion capture and recognition accuracy based on extension immune neural network has greater advantages than the classical immune algorithm.**

Keywords—extension analysis, immune neural network (INN), sign language key actions, hu invariant moment characteristics, Hamming rules

I. INTRODUCTION

In daily communication, people not only speak by mouth, but also transmit information by gesture language. As a special gesture language, sign language is essential for the hearing impaired. Unfortunately, the number of healthy people who are proficient in sign language in daily life is relatively small [1]. Most of the healthy people know nothing about sign language except for a few people who are engaged in special education, and most people will not take the initiative to spend time and energy to learn sign language. Besides, there are few professional sign language institutions in the current education market, and the sign language translation major is rarely engaged in [2]. That healthy people can not communicate with the deaf normally, which makes the social environment unable to be adapt well by deaf. Therefore, it has profound research significance and good social impact to study sign language recognition technology and create a barrier free communication platform between deaf and healthy people [3]. Sign language recognition technology can help healthy people to enter the world of the deaf and realize their inner emotional needs. It can also help deaf people better integrate into the contemporary society and enhance their sense of social belonging and life happiness [4].

Visual based sign language recognition is a significant and practical research topic, but it is also very difficult. Research on sign language recognition technology can not only promote the development of human-computer interaction, but also help the deaf better participate in social life [5]. Until today, vision-based sign language recognition technology has obtained a lot of research results, but most of the current research is aimed at the study of simple sign language vocabulary, and the study of sign language sentence recognition is relatively less [6]. In the research process of sign language sentence recognition, there are still some problems and challenges in how to effectively segment the sentence action boundary and improve the accuracy and speed of recognition.

The sign language recognition technology based on computer vision refers to taking sign language video by camera and analyzing and processing the sign language video by algorithm to complete the recognition. Because the monocular camera can only obtain two-dimensional information, which is difficult to capture the changes of gesture in three-dimensional space [7]. However, positioning with color gloves is not in line with the natural human-computer interaction criteria [8]. Positioning with skin color will lead to confusion with facial skin, so researchers try to use multi-eye cameras to obtain more information [9].

With the introduction of Kinect, the study of sign language recognition has entered a new stage. It is not only cheap and easy to carry, but also can detect the depth information of the collected image in real time, which makes the tracking of the hand position and the detection of the hand area very simple [10]. It promotes the research of sign language recognition technology based on computer vision.

In order to solve the problem of key action extraction of sign language, the method of target extraction based on artificial immune is adopted in this paper. The ability of self-learning, self-dynamic adaptation and feedback of the artificial immune system is very sound, and it has strong self-adaptive, self-learning and "self" and "non self" recognition ability. Immune neural network refers to the application of the topological structure of immune system in the field of engineering [11]. With the help of abundant activities among antibodies and antigens of immune system, communication, contact and restriction within the network can be realized. It can not only combine with specific antigen to play the role of antibody, but also trigger the immune response with its own unique antigenic determinant [12]. A dynamic and balanced network structure is formed by self- identification, mutual stimulation and mutual restriction.

Based on the progress made by existing research, this paper improves the pattern matching and pattern recognition

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algorithm of immune neural network, and uses it to extract the target area of Chinese sign language key actions accurately, enriches the existing research content of sign language recognition, and improves the accuracy of sign language detection and recognition based on artificial immune algorithm.

The structure of the following parts of this paper is as follows: the second part introduces the core content of immune neural network and related algorithm improvement, the third part introduces the design of immune detector, the fourth part introduces the simulation experiment results and analysis, the fifth part is the summary of this paper and the future work prospects.

II. Sign language action extraction algorithm based on Artificial Immune Network

According to the topological structure of the immune response mechanism of the biological immune system, that is namely: the antigens are engulfed by B cells and appear on the surface of B cells, antigen presenting cells are found by helper T cells, and secretion of interleukin (IL +) activates the immune response. IL + becomes the second signal of B cells, and the stimulated B cells to decompose into plasma cells, then synthesize and secrete antibodies; if antibodies increase, inhibit T cells from secreting interleukin (IL-) after stimulation [13].

We designed the immune neural network for sign language target detection as shown in Figure 1: In this model, the antigen is used as the input and the antibody is used as the output. The output depends on the interaction between B cells and T cells, not just by B cell decision; the inhibition of T cells is very important in the immune system; the memory mode is used as the connection weight of auxiliary T cells and B cells, and the antibody is used as the error between the input mode and the memory mode [14]. The network combines the biological immune system and mimics the mechanism of interaction between B and T cells of the immune system. The results show that the model not only has good memory ability, but also can effectively suppress noise.

It is worth noting that in the mode space where the image is located, there will always be areas that cannot be covered by the detector. Artificial immune theory calls these areas "black holes". Due to the existence of "black holes", there will always be a certain fuzzy set S (as shown in Figure 2) that cannot be identified. Traditional immune neural networks cannot effectively solve the problem of "black hole". In this paper, Hamming rules are introduced to match action features, and feature detectors are designed with scale extension, which effectively improves the "black hole" coverage and improves the accuracy of feature extraction in the pattern space where the image is located.

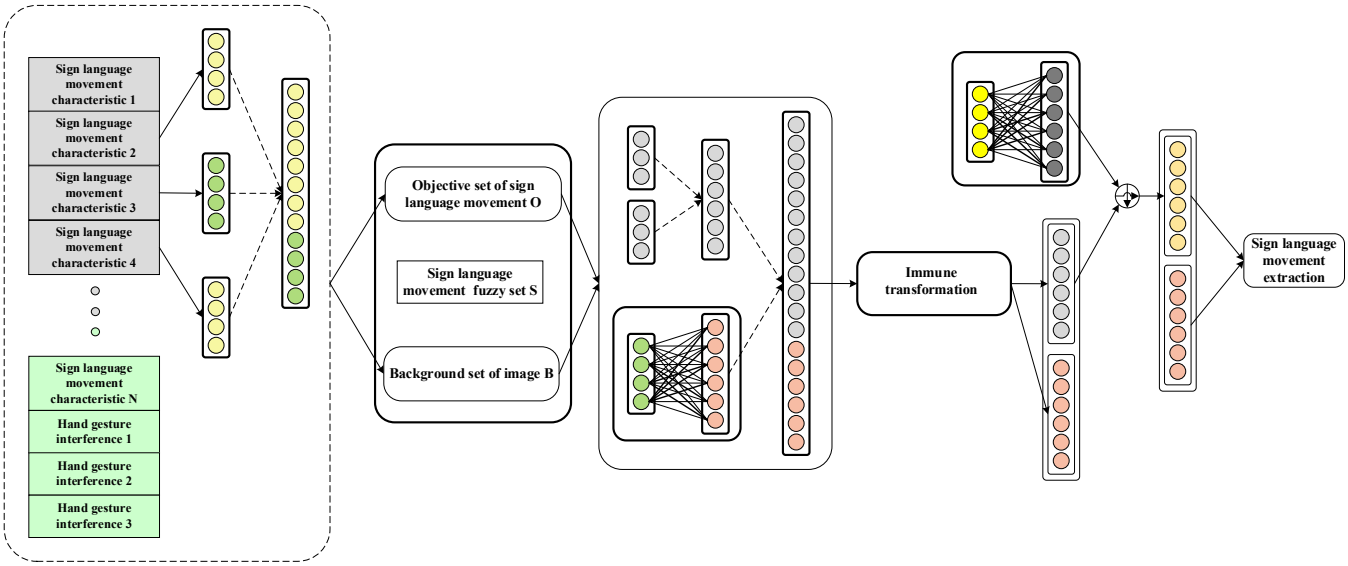


Fig. 1. Immune neural network sign language action feature extraction flow chart

The core idea of the scale-adaptive extension detector design based on Hamming rules is to use the matching method of Hamming rules to reduce the number of limited detectors required to meet the requirements of Hamming rule matching failure as much as possible. Adaptive in a fixed area reduces the number of "black holes" in the system as much as possible, thereby effectively increasing the space covered by the effective detectors produced by the system.

Two pattern character strings of length a, when they have the same character in no less than r identical positions (that is, the Hamming distance of the two strings is not greater than a-r), it is said to match under the Hamming rules. The probability of matching is:

$$P_{M-h} \approx 2^{-a} \sum_{i=r}^{a} \frac{a!}{(a-i)!i!} \tag{1}$$

In the above formula, $\frac{a!}{(a-i)!i!}$ represents all possibilities of taking any r-bit C_a^i.

The extension detector is constructed as follows: defined P_M as the matching degree between two pixels in two spaces, N_R is the number of effective detectors, and P_f is the matching failure rate, then the probability that each effective detector does not detect the change in matching degree is $(1-P_M)$,the probability N_R that the effective detector does

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not detect the element change is:

$$P_f = (1 - P_M)^{N_R} \tag{2}$$

The probabilities of element changes are organized as:

$$N_R = \frac{\lg P_f}{\lg(1 - P_M)} \tag{3}$$

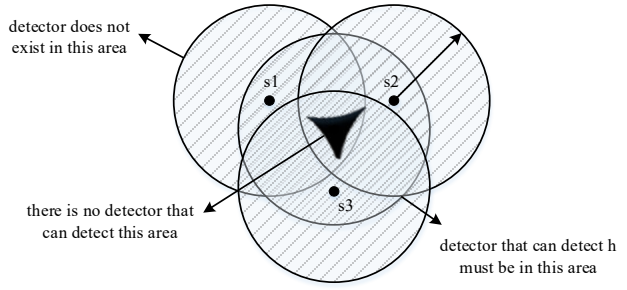


Fig. 2. Diagram of black hole in image mode space

In the process of target location and extraction of key action areas of sign language, in order to meet the accuracy requirements, the matching failure rate should be reduced to less than 0.01.

III. Action Feature Recognition Of Sign Language Based On Extenion Immune Network

The input of the network is the sign language scene image. Under the function of each link in the network, the recognition result of the sign language action segmentation is finally output. The key feature extraction layer of sign language locates, collected, and filtered, and the sign language. The feature matching layer is used to classify the sign language actions, and then sent to the feature recognition layer for key action recognition. The overall working process of the system is shown in Figure 3.

IV. Experiment And Result Analysis

The experimental environment in this paper is shown in Table 1. All experiments in this paper are carried out in this experimental environment.

First, four basic variables are defined: TP (true positive): the total number of pixels correctly classified as foreground; TN (true negative): the total number of pixels correctly classified as background; FP (false positive): the total number of pixels incorrectly classified as foreground; FN (false negative): the total number of pixels incorrectly classified as background. *Recall* is the detection accuracy, and the formula is as follows:

$$Recall = \frac{TP}{TP + FN} \tag{4}$$

Its value is between 0-1. The closer it is to 1, the better the detection effect is. The results of relevant data are shown in Figure 4.

TABLE I. Experimental Environment

Environment	Explanation	Model
Hardware environment	CPU	Intel i5-6200
	memory	4GB
	graphics card	128MB
Software environment	operating system	Windows 10
	development tool	MATLAB2016b
	development language	MATLAB

According to the experimental results, as shown in Figure 5, for the sample images in the video sequence of sign language used in the experiment, the algorithm in this paper can effectively locate the sign language region and extract the key features.

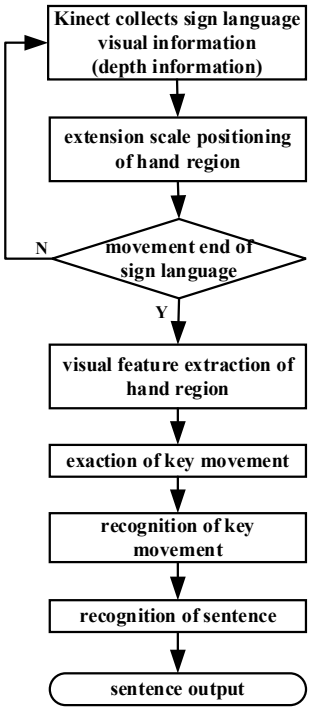


Fig. 3. System work flow chart

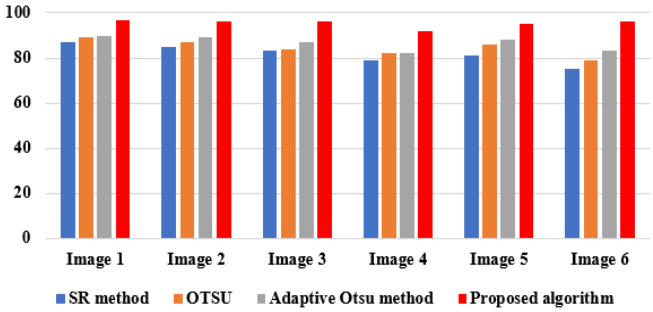


Fig. 4. Comparison of detection accuracy of different algorithms

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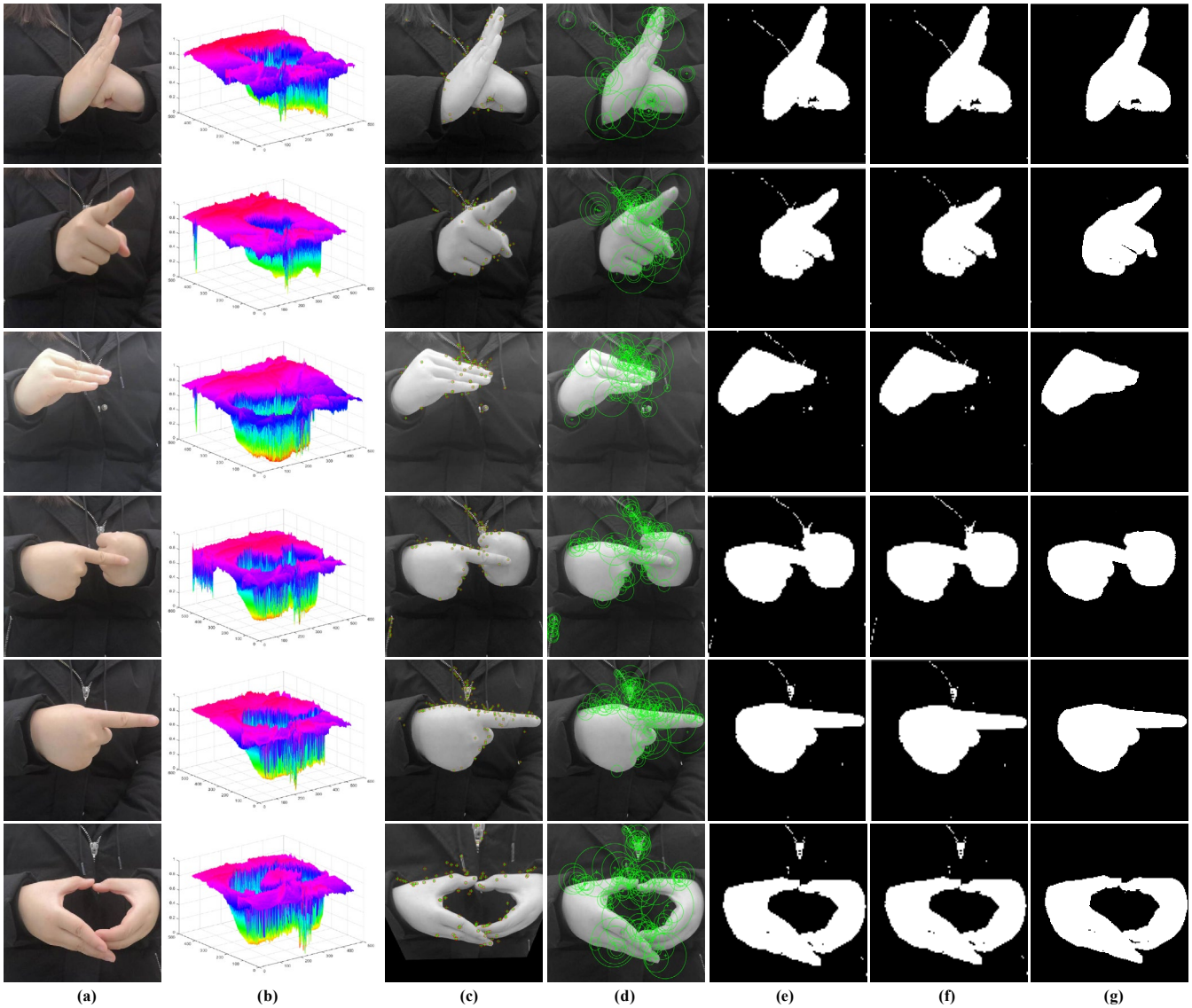


Fig. 5. Comparison of target detection and extraction methods; (a) Original images; (b) Three dimensional energy distribution diagrams; (c) Immune detectors; (d) Working process of detectors; (e) SR method; (f) OTSU; (g) Proposed algorithm.

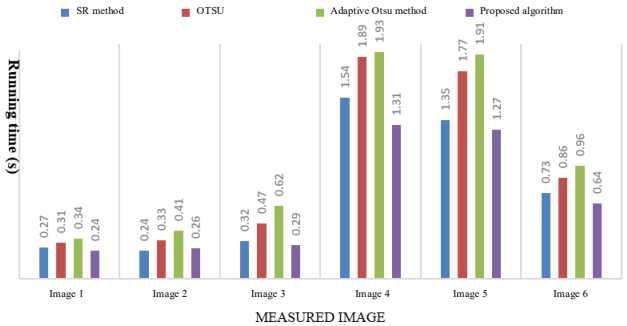


Fig. 6. Comparison of running time of different algorithms

V. Conclusion

In this paper, the basic idea of using key actions to describe continuous sign language is proposed, and the extension analysis theory is introduced into computer vision technology to take key actions as primitives of sign language sentences, which greatly reduces the complexity of research. The recognition of continuous sign language sentences is transformed into the recognition of discrete action sequences, which reduces the difficulty of recognition. According to the problem that hand movement leads to low positioning accuracy in sign language interaction, a scale extension detector is proposed to accurately locate the hand area. Combined with the key action classification algorithm of the extension immune neural network, the key action detection and recognition are realized.

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